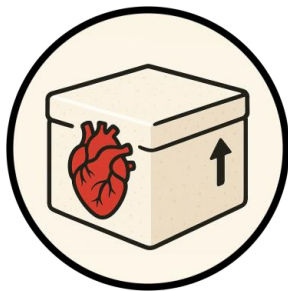


The Cooler Heart

A heart transplant shipping device



Darcey Kate Britt

Anna Claire Gooding



Paul
Specht



Jacob Louie



Jared
Kasten



**What is the leading cause
of death in the US?**

Heart Disease

680,981



deaths in 2023



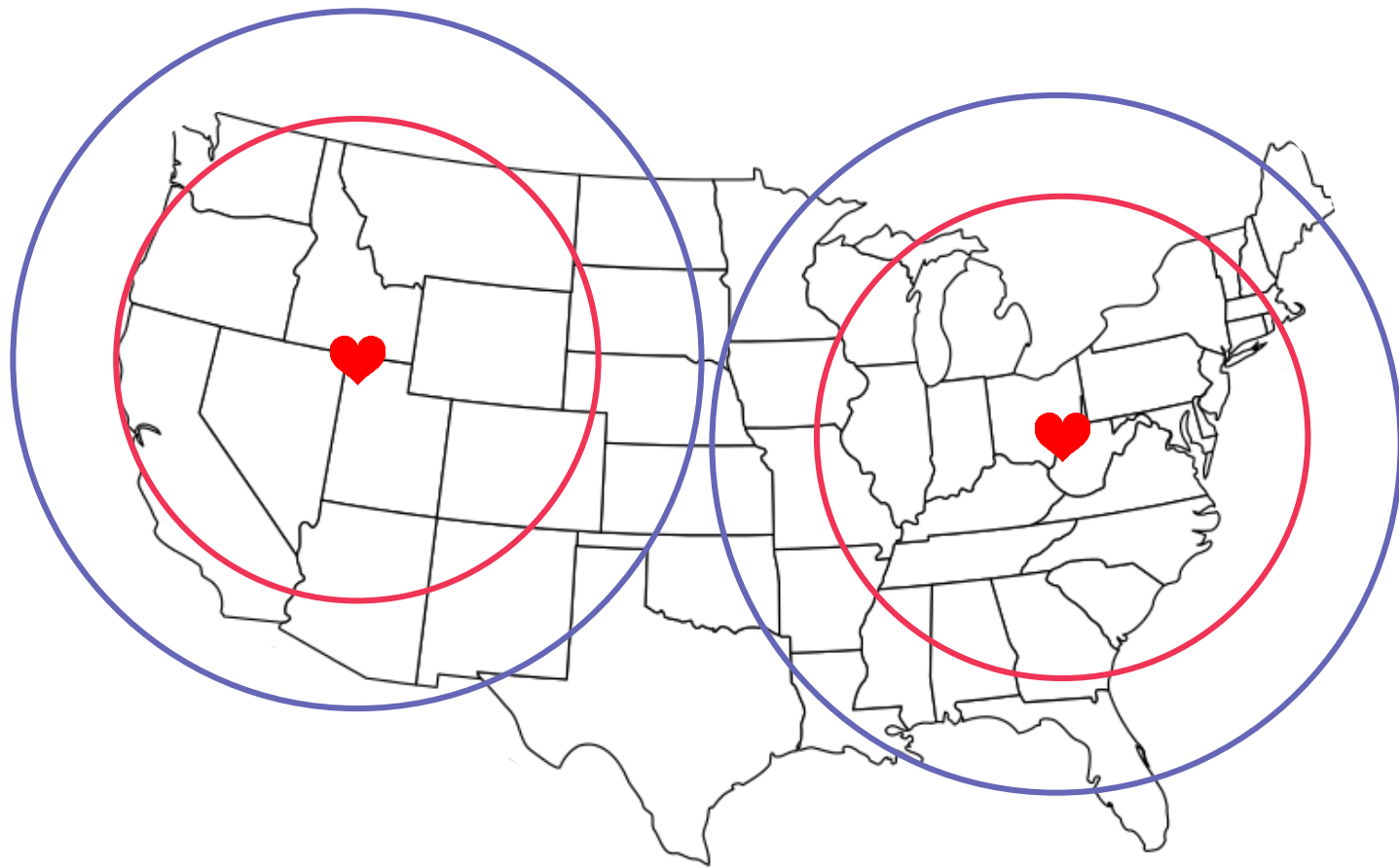
3,244 on transplant list

The Problem



The **current standard** of heart preservation is **limited to 4 hours**





1 more hour increases radius to thousands more potential candidates

Current Solutions



Classic Ice Cooler

Plastic bags in an ice box

Uneven cooling



SherpaPak

Hypothermic preservation

Current US standard of
care

Key Innovations



Heart preserved at 4-8°C



Perfusion provides **nutrients** and **oxygen**



Displays temperature, flow rate, oxygen concentration



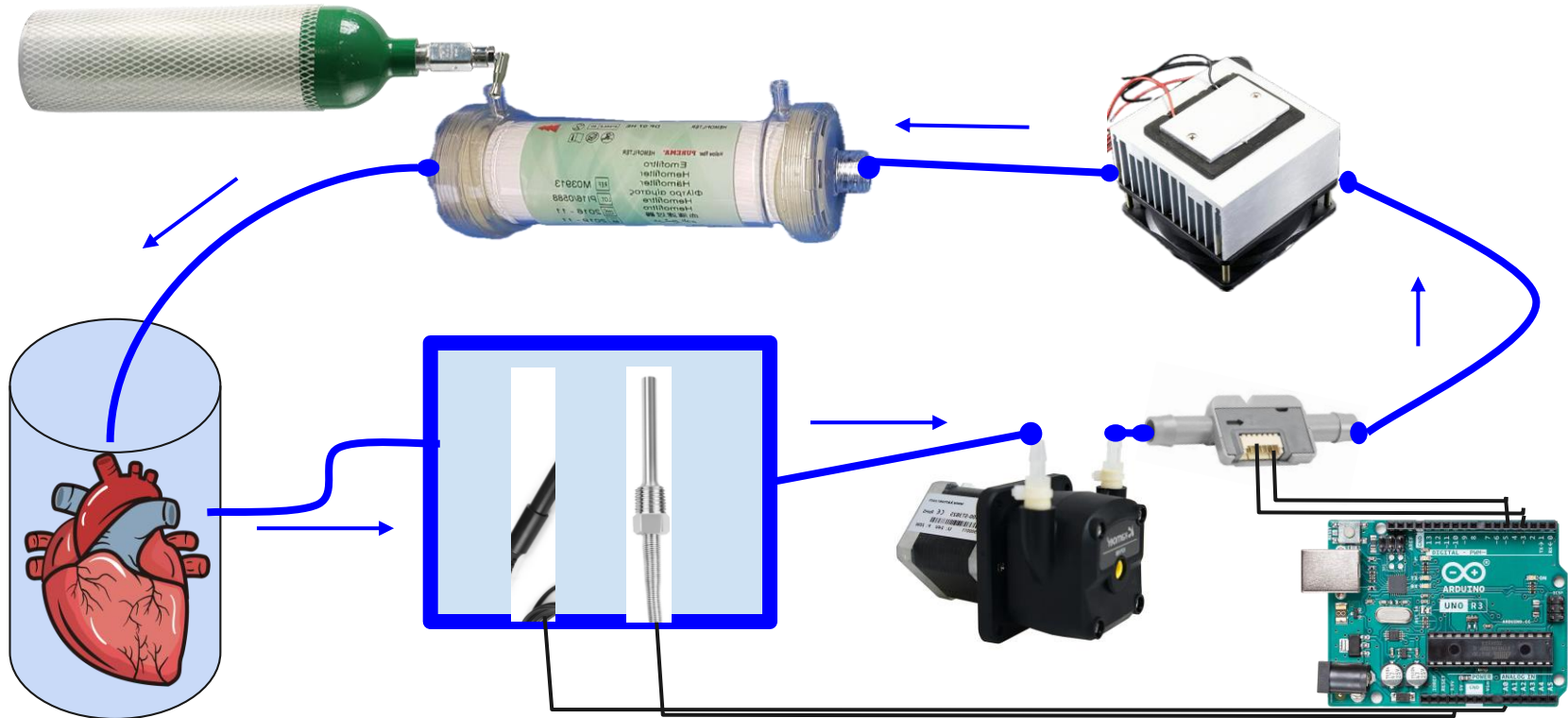
Efficient **workflow** that maintains **sterility**

The Idea

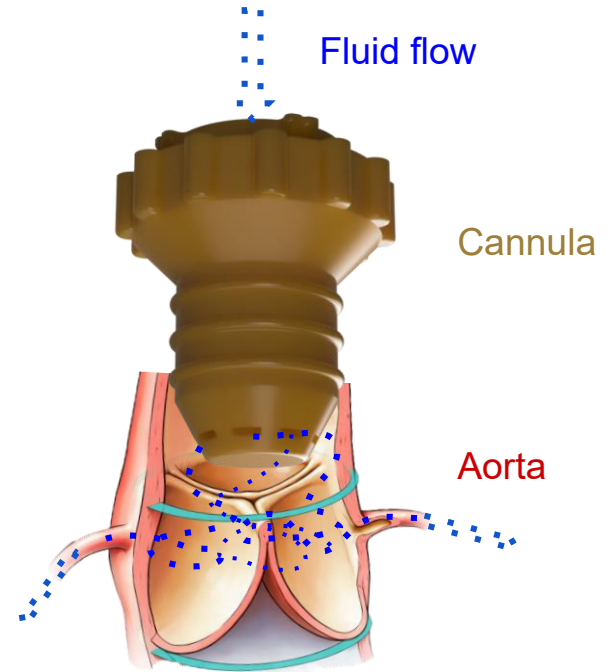
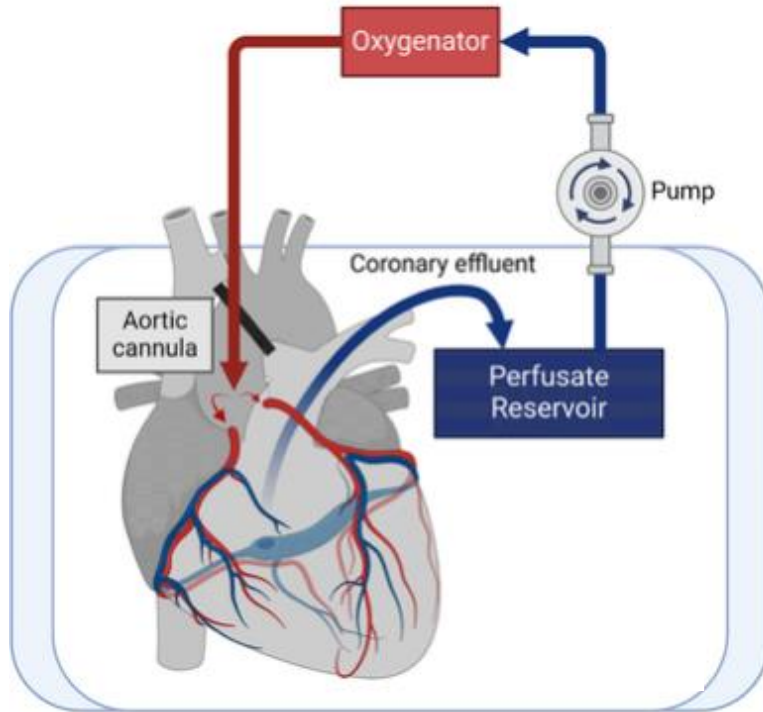
Hypothermic oxygenated perfusion



The Circuit



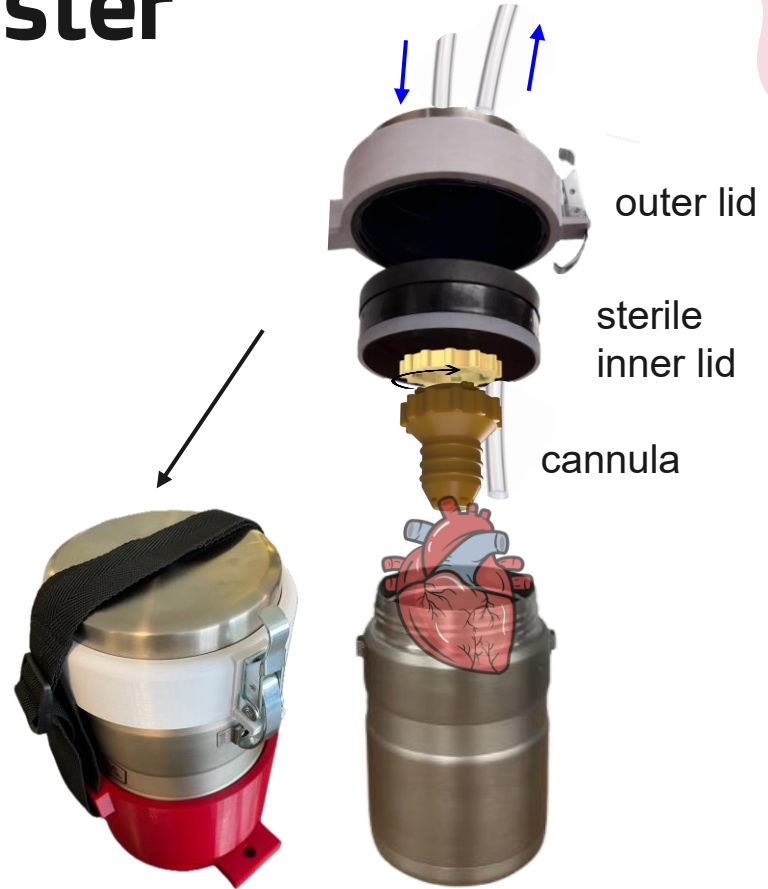
Aortic Cannulation



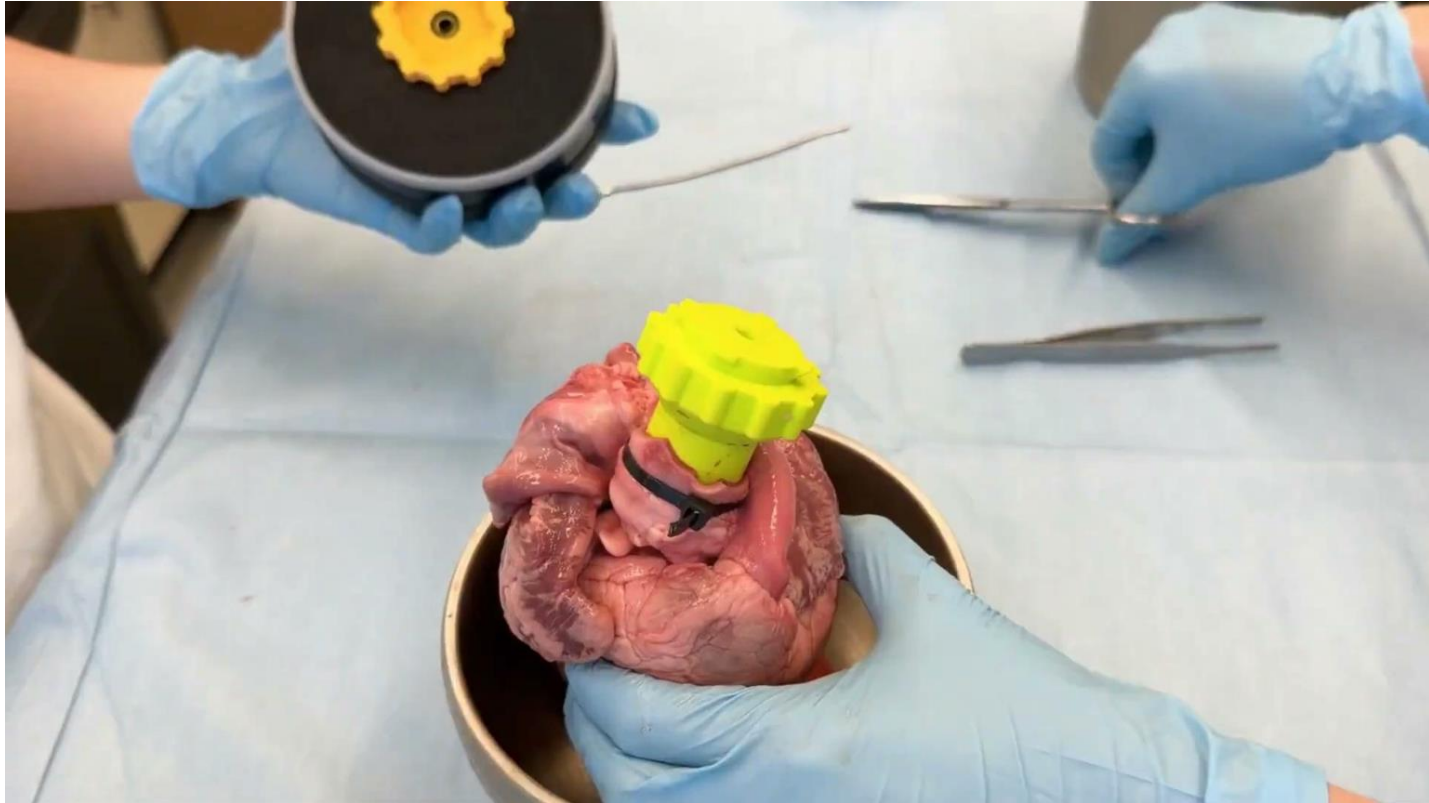
Heart Canister

Double layer sterility

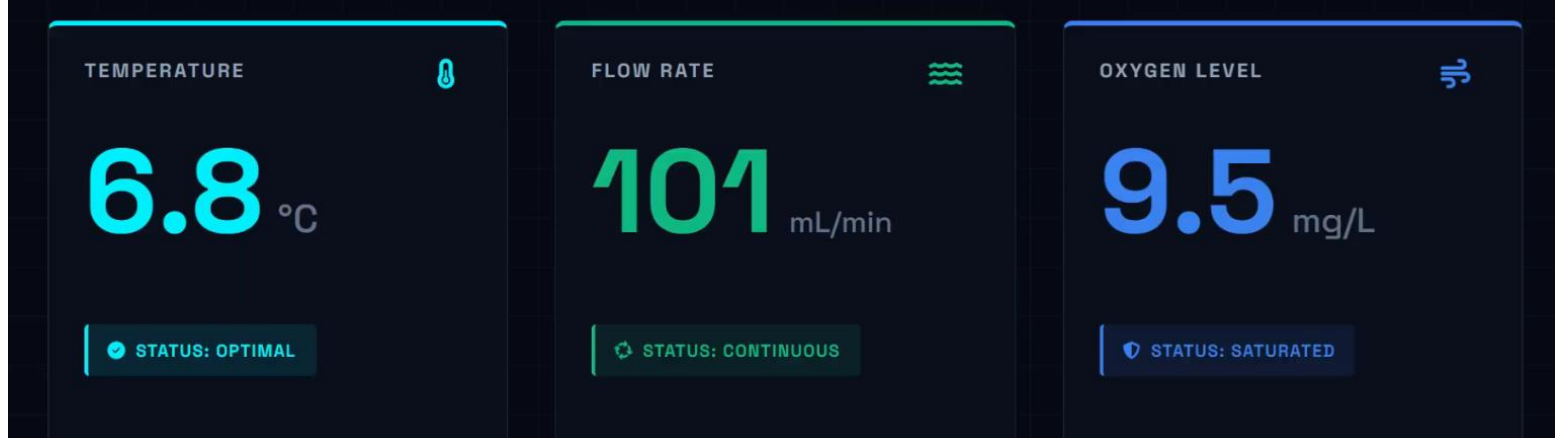
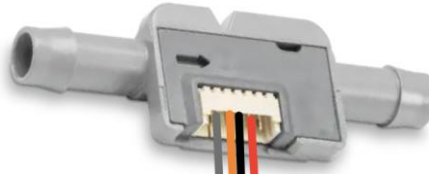
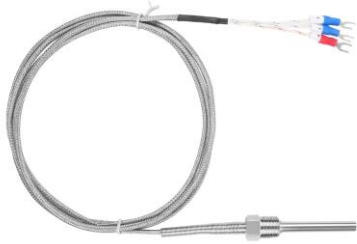
Attachment of heart to device



Set Up



Monitoring in Transport



Arrival



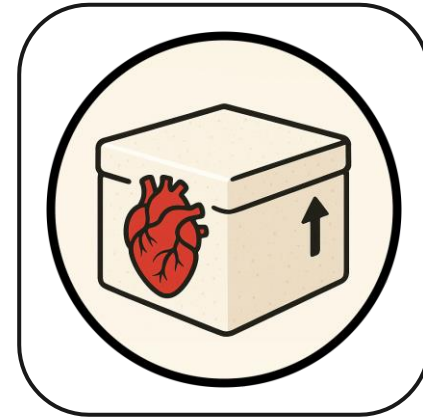
Accomplishments

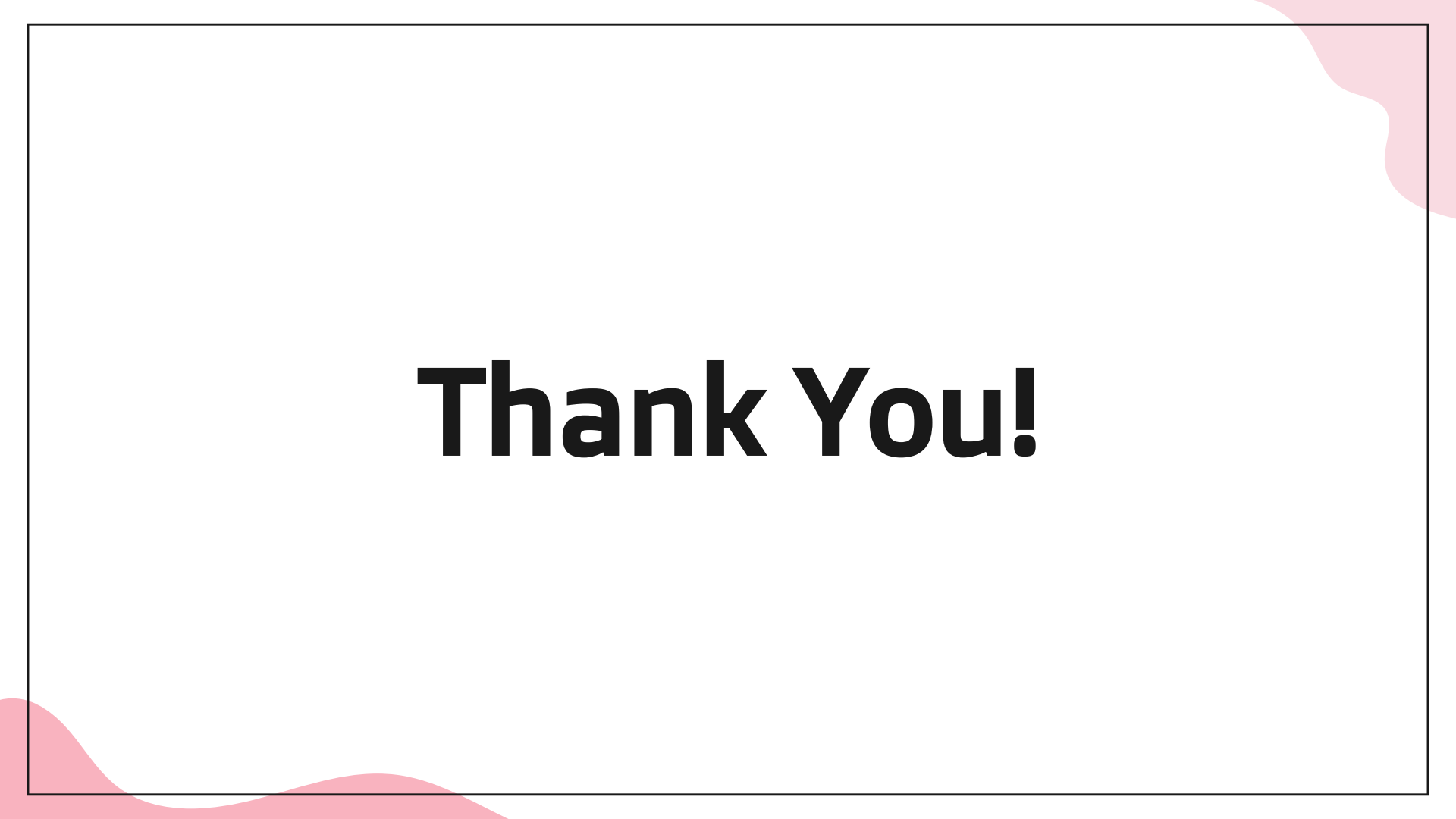
Data Monitoring



Future Development

- Potential viability metrics: lactate concentration, calcium measurements
- A Cooler Heart app



The image features a white background with a thin black rectangular border. In the top-right and bottom-left corners, there are decorative, wavy pink shapes that extend slightly beyond the border. Centered in the middle of the frame is the text "Thank You!" in a large, bold, black sans-serif font.

Thank You!